

A Step-By-Step Guide to Implement AWS CI/CD Pipeline

Lets now build a CI/CD Pipeline with AWS by detailing step by step as follows:

Step 1: Create IAM Role for EC2 and AWS CodeDeploy

- Navigate to [IAM](#) service.
- Then go to roles and create a new role.
- Select trusted entity type as **AWS Service** and use case as [EC2](#)

The screenshot shows the 'Select trusted entity' page in the AWS IAM console. It has a dark theme. At the top, it says 'Select trusted entity' with an 'Info' link. Below this is a section titled 'Trusted entity type' with five options, each in a box with a radio button: 'AWS service' (selected), 'AWS account', 'Web identity', 'SAML 2.0 federation', and 'Custom trust policy'. Each option has a brief description. Below this is a 'Use case' section with the text 'Allow an AWS service like EC2, Lambda, or others to perform actions in this account.' and a dropdown menu labeled 'Service or use case' with 'EC2' selected. Below the dropdown, it says 'Choose a use case for the specified service.' and 'Use case'. There are two options with radio buttons: 'EC2' (selected) and 'EC2 Role for AWS Systems Manager'. The 'EC2' option has a description: 'Allows EC2 instances to call AWS services on your behalf.'

Step 2: Add permissions To IAM Role

- Select **AmazonS3ReadOnlyAccess** permission. It will allow our EC2 instance to access stored artifacts from the Amazon S3 bucket.

The screenshot shows the 'Add permissions' page in the AWS IAM console. It has a dark theme. At the top, it says 'Add permissions' with an 'Info' link. Below this is a section titled 'Permissions policies (1/884)' with an 'Info' link and the text 'Choose one or more policies to attach to your new role.' There is a search bar with 'AmazonS3ReadOnlyAccess' entered and a 'Filter by Type' dropdown set to 'All types'. Below this is a table with columns 'Policy name' and 'Type'. The table has one row with a checkbox, a plus icon, the policy name 'AmazonS3ReadOnlyAccess', and the type 'AWS managed'. At the bottom, there is a link '► Set permissions boundary - optional'.

Step 3: Creating The Role For AWS CodeDeploy

- Provide the Name, review and Click on Create for creating the Role.
- Select an appropriate role name and click on create role.

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

EC2RoleAWSCodeDeploy-1

Maximum 64 characters. Use alphanumeric and '+', '=', '@', '-', '_' characters.

Description
Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use alphanumeric and '+', '=', '@', '-', '_' characters.

Step 4: Creating New Service Role For CodeDeploy

- Create a new service role for CodeDeploy and attach AWSCodeDeployRole policy which will provide the permissions for our service role to read tags of our EC2 instance, publish information to Amazon SNS topics and much more task.
- Repeat the Above 3 steps again with trusted entity type **AWS Service**, use case [CodeDeploy](#).

Select trusted entity Info

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

CodeDeploy

Choose a use case for the specified service.

Use case

☒ **CodeDeploy**
Allows CodeDeploy to call AWS services such as Auto Scaling on your behalf.

☐ **CodeDeploy for Lambda**
Allows CodeDeploy to route traffic to a new version of an AWS Lambda function version on your behalf.

☐ **CodeDeploy - ECS**
Allows CodeDeploy to read S3 objects, invoke Lambda functions, publish to SNS topics, and update ECS services on your behalf.

- Add **AWSCodeDeployRole** permissions to this creating Role

Add permissions [Info](#)

Permissions policies (1) [Info](#)
The type of role that you selected requires the following policy.

Policy name ↗	Type
AWSCodeDeployRole	AWS managed

► **Set permissions boundary - optional**

- Provide the Name, review and create the role.

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+=, @-_' characters.

Step 5: Launch An Linux EC2 instance

- Select the instance with AMI such as "**Amazon Linux**" and connect to CLI Console.
- Switch to **root user** from **ec2-user** to gain admin access power by using following command "**sudo su**" in Linux.

```
sudo su
```

Step 6: Update The Packages

- The command "sudo yum update" is used in Amazon Linux, [CentOS](#), and Red Hat Linux distributions to update installed packages on your system to their latest available versions.

```
sudo yum update
```

Step 7: Install The Ruby And Wget Software

- The command '**sudo yum install ruby**' is used to install the Ruby programming software using the YUM package manager.

```
sudo yum install ruby
```

- The command `sudo yum install wget` is used to install the "wget" package on a system running Amazon Linux, CentOS, or other Red Hat-based Linux distributions that use the YUM package manager.

```
sudo yum install wget
```

Step 8: Download CodeDeploy Agent Script

- Downloading the AWS CodeDeploy agent installation script from the [AWS S3](https://aws-codedeploy-us-east-1.s3.amazonaws.com/latest/install) bucket is an essential step in setting up AWS CodeDeploy for your infrastructure.
- The CodeDeploy agent is a lightweight, scalable software component that enables AWS CodeDeploy to deploy and manage applications on your EC2 instances or on-premises servers.

```
wget https://aws-codedeploy-us-east-1.s3.amazonaws.com/latest/install
```

Step 9: Run Installation Script

- The command `chmod +x ./install` is used to make a file executable in a Unix-like operating system, including Linux.

```
chmod +x ./install
```

The command '**sudo ./install auto**' is likely used to run an installation script with superuser (administrator) privileges and pass the "auto" argument to the script.

```
sudo ./install auto
```

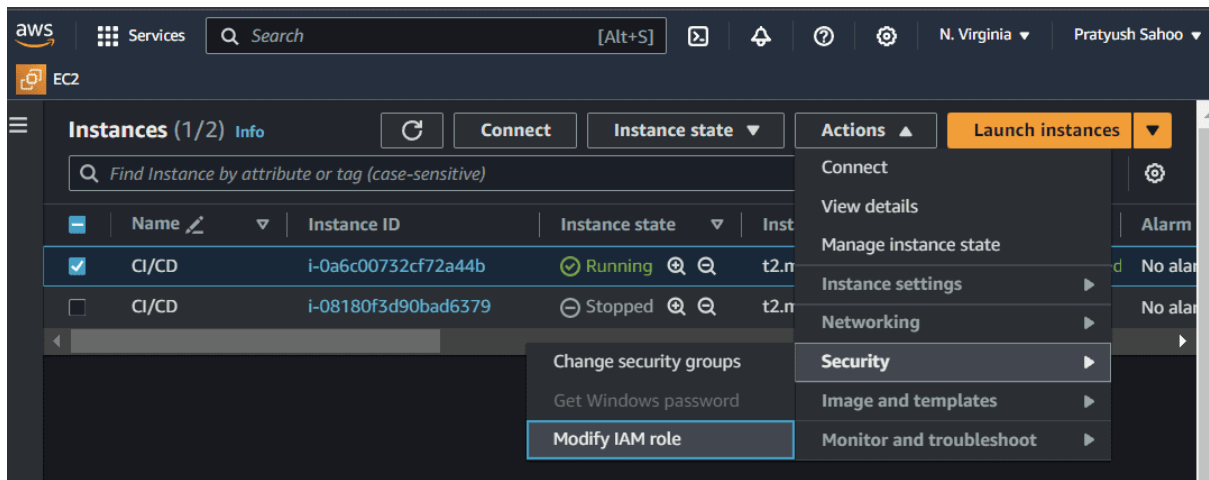
Step 10: Check CodeDeploy Agent Status

- The command `sudo service codedeploy-agent status` is used to check the status of the AWS CodeDeploy agent running on your system.

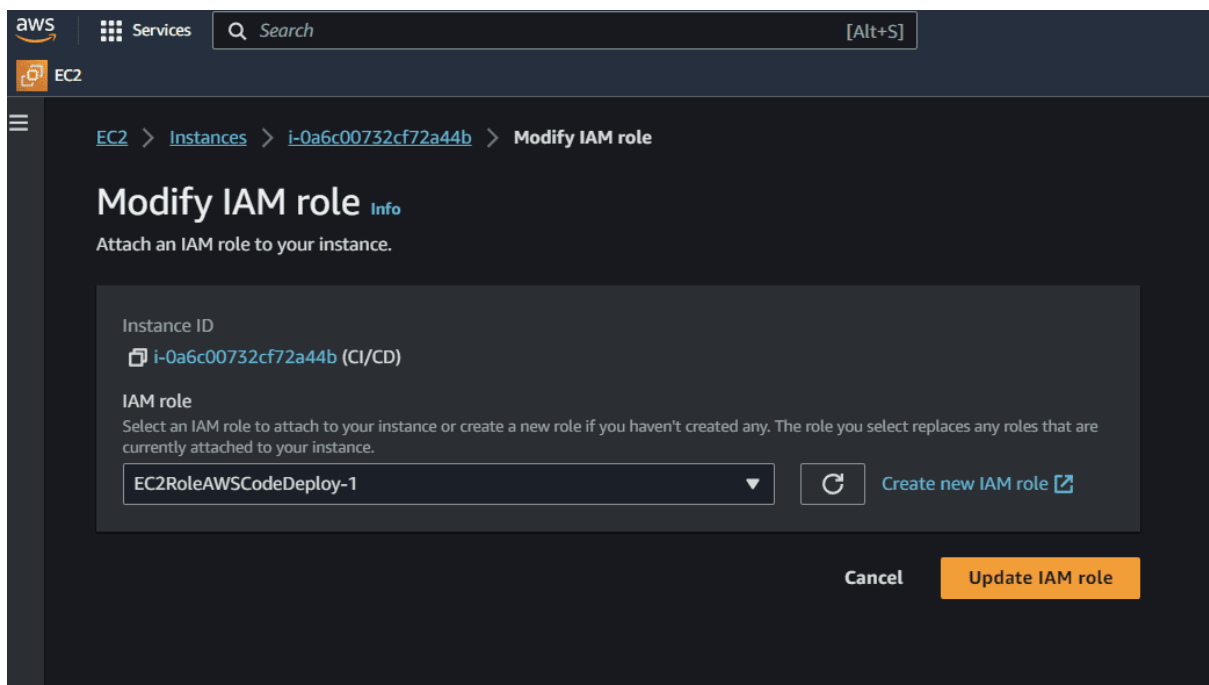
```
sudo service codedeploy-agent status
```

Step 11: Modifying IAM Role

- After running the following commands, select the instance and click on "Actions", then click on "Security" and click on "Modify IAM Role". Then choose the above created IAM Role and click on "Update IAM Role".
- After this step, your EC2 instance gets attached with your above created IAM Role.



- Modify the IAM role by clicking on the button **Update IAM role** as shown in the figure.



Step 12: Finalizing The Configuration

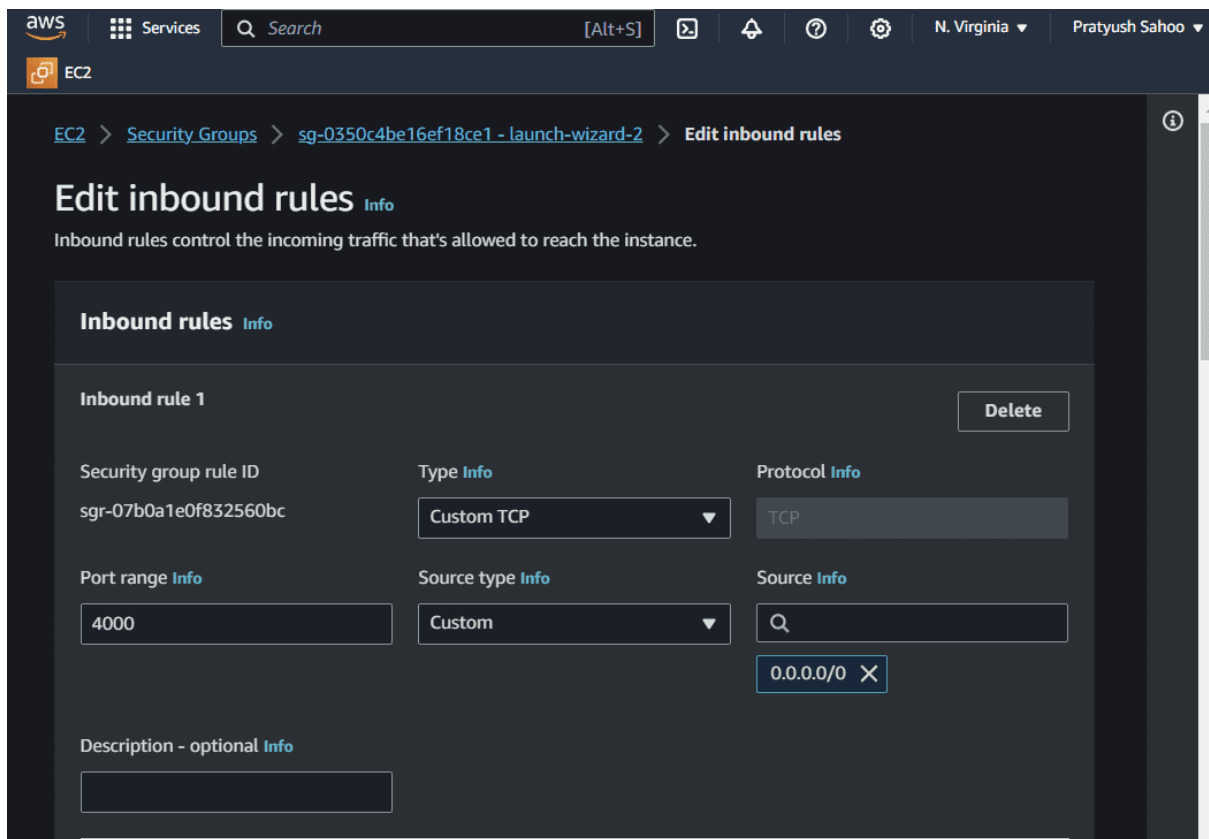
After this process, go to the console where your instance is connected and run the command "exit" to exit from the root folder and go back to the EC2 folder. Make a directory on the EC2 folder named "server", this is the directory where my source code will be deployed.

The screenshot shows an AWS Management Console terminal window for an EC2 instance. The terminal output indicates that the AWS CodeDeploy agent has been successfully installed and is running as PID 30588. The user has also created a directory named 'server' and is ready to install the application. The terminal output is as follows:

```
Complete!
I, [2023-10-22T18:30:38.357273 #30431] INFO -- : Update check complete.
I, [2023-10-22T18:30:38.357450 #30431] INFO -- : Stopping updater.
[root@ip-172-31-27-184 ec2-user]# sudo service codedeploy-agent status
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modules/global_help_format.rb:37: warning: Passing safe_level with the 2nd argument of ERB.new is deprecated. Do not use it, and specify other arguments as keyword arguments.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modules/global_help_format.rb:37: warning: Passing trim_mode with the 3rd argument of ERB.new is deprecated. Use keyword argument like ERB.new(str, trim_mode: ...) instead.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modules/command_help_format.rb:27: warning: Passing safe_level with the 2nd argument of ERB.new is deprecated. Do not use it, and specify other arguments as keyword arguments.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modules/command_help_format.rb:27: warning: Passing trim_mode with the 3rd argument of ERB.new is deprecated. Use keyword argument like ERB.new(str, trim_mode: ...) instead.
The AWS CodeDeploy agent is running as PID 30588
[root@ip-172-31-27-184 ec2-user]# ls
install
[root@ip-172-31-27-184 ec2-user]# exit
exit
[ec2-user@ip-172-31-27-184 ~]$ ls
install
[ec2-user@ip-172-31-27-184 ~]$ mkdir server
[ec2-user@ip-172-31-27-184 ~]$ ls
install server
[ec2-user@ip-172-31-27-184 ~]$
```

Below the terminal window, the instance ID **i-0a6c00732cf72a44b (CI/CD)** is displayed, along with its PublicIPs (54.163.18.150) and PrivateIPs (172.31.27.184).

- Then after doing the above process, come back to the running instances list.
- Select your currently created running instance and go to the "Security" section present at the end of the page.
- Click on the link present under the "Security Groups". After redirecting to the required page, click on "Edit Inbound rules" under the section of "Inbound rules" present at the end of the page.
- Then add a rule, select a port range of your choice and select the source as "Anywhere-IPv4" from the dropdown menu and then click on "Save rules".
- Basically, let me give you a overview what we are actually doing here. In brief, when you add an inbound rule to a security group for an instance with port range (in my case, it was 4000) and set the source to "Anywhere-IPv4," you are allowing any computer or device on the internet to connect to your instance through port 4000.
- This is like opening a door (port 4000) on your server and letting anyone from anywhere access the service or application running on that port.



Step 13: Create A New Pipeline

- Create a CodePipeline using Github, CodeBuild and CodeDeploy
- Firstly Create CodePipeline navigate to CodePipeline via AWS Management Console and click on Create pipeline.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Choose pipeline settings Info

Pipeline settings

Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.

reactDemoGithubConnection-1

No more than 100 characters

Service role

☒ **New service role**
Create a service role in your account

☐ **Existing service role**
Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-us-east-1-reactDemoGithubConnection-

Type your service role name

☒ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

► **Advanced settings**

Cancel Next

Step 14: Choose Github In Code Source

- After selecting GitHub as the source provider, click on the Connect to GitHub button. You'll then be prompt to enter your GitHub login credentials.
- Once you grant AWS CodePipeline access to your GitHub repository, you can select a repository and branch for CodePipeline to upload commits to this repository to your pipeline.

Step 2

Add source stage

Step 3

Add build stage

Step 4

Add deploy stage

Step 5

Review

Source

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2)

New GitHub version 2 (app-based) action

To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. [Learn more](#)

Connection

Choose an existing connection that you have already configured, or create a new one and then return to this task.

Q

arn:aws:codestar-connections:us-east-1:160990010622:connection/9757e0f

X

or

Connect to GitHub

Ready to connect

Your GitHub connection is ready for use.

Repository name

Choose a repository in your GitHub account.

Q

DotUrDesign/codepipelinedemo

X

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.

Branch name

Choose a branch of the repository.

Q

main

X

Change detection options

☒ Start the pipeline on source code change

Automatically starts your pipeline when a change occurs in the source code. If turned off, your pipeline only runs if you start it manually or on a schedule.

Output artifact format

Choose the output artifact format.

☒ CodePipeline default

AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.

☐ Full clone

AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions.

Cancel

Previous

Next

Step 15: Configure CodeBuild (Optional)

- If you haven't created a project prior to creating your pipeline, then you can create a project directly from here by clicking Create project button.
- Note: Buildspec file is a collection of build commands and related settings, in YAML format, that CodeBuild uses to run a build. For my project, I created a buildspec.yaml file and added it in the root of my project directory.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Add build stage Info

Build - *optional*

Build provider
This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

AWS CodeBuild

Region
US East (N. Virginia)

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

reactDemoBuildProject or [Create project](#)

Environment variables - *optional*
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

[Add environment variable](#)

Build type

☒ **Single build**
Triggers a single build.

☐ **Batch build**
Triggers multiple builds as a single execution.

Step 16: Add Deploy Stage

Note : Before going to configure Add Deploy Stage, Let's make duplicate tab of current tab.

- Go to code deploy in the navigation, Select Application, then add create a deployment group.

Developer Tools > CodeDeploy > Applications > Create application

Create application

Application configuration

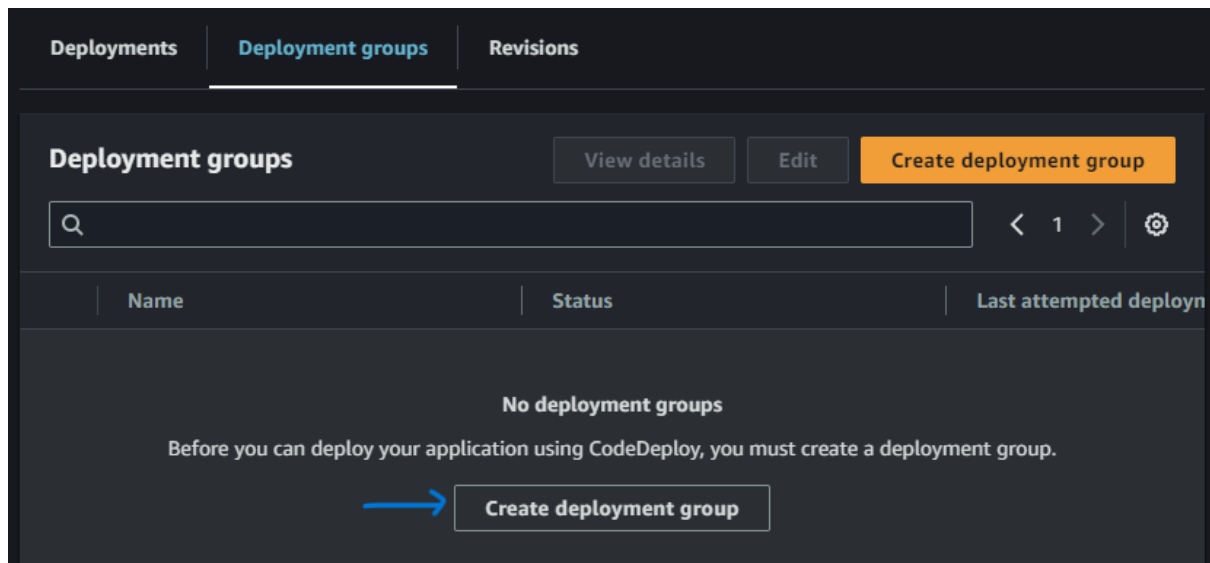
Application name
Enter an application name
ec2App
100 character limit

Compute platform
Choose a compute platform
EC2/On-premises

Tags
[Add tag](#)

[Cancel](#) [Create application](#)

- Create a deployment Group by clicking on the button "**Create deployment group**", the following screenshot illustrates with practically.



- In deployment group Select EC2 instances and select Tag and Value

Create deployment group

Application

Application
ec2App
Compute type
EC2/On-premises

Deployment group name

Enter a deployment group name

AppDepGrp

100 character limit

Service role

Enter a service role

Enter a service role with CodeDeploy permissions that grants AWS CodeDeploy access to your target instances.

Q arr:aws:iam::160990010622:role/AWSCodeDeployRole-1



Deployment type

Choose how to deploy your application

☒ **In-place**

Updates the instances in the deployment group with the latest application revisions. During a deployment, each instance will be briefly taken offline for its update

☐ **Blue/green**

Replaces the instances in the deployment group with new instances and deploys the latest application revision to them. After instances in the replacement environment are registered with a load balancer, instances from the original environment are deregistered and can be terminated.

- Provide the Environment configurations such as select the **Amazon EC2 Instances** and provide the key and values to it.

Environment configuration

Select any combination of Amazon EC2 Auto Scaling groups, Amazon EC2 instances, and on-premises instances to add to this deployment

☐ Amazon EC2 Auto Scaling groups

☒ Amazon EC2 instances

2 unique matched instances. [Click here for details](#)

You can add up to three groups of tags for EC2 instances to this deployment group.

One tag group: Any instance identified by the tag group will be deployed to.

Multiple tag groups: Only instances identified by all the tag groups will be deployed to.

Tag group 1

Key

Value - optional

Q Name



Q CI/CD



Remove tag

Add tag

+ Add tag group

☐ On-premises instances

Matching instances

2 unique matched instances. [Click here for details](#)

Agent configuration with AWS Systems Manager [Info](#)



Complete the required prerequisites before AWS Systems Manager can install the CodeDeploy Agent. Make sure the AWS Systems Manager Agent is installed on all instances and attach the required IAM policies to them. [Learn more](#)

Install AWS CodeDeploy Agent

☐ Never

☐ Only once

☒ Now and schedule updates

- Uncheck Load Balancer Option

Agent configuration with AWS Systems Manager [Info](#)



Complete the required prerequisites before AWS Systems Manager can install the CodeDeploy Agent. Make sure the AWS Systems Manager Agent is installed on all instances and attach the required IAM policies to them. [Learn more](#)

Install AWS CodeDeploy Agent

- ☐ Never
- ☐ Only once
- ☒ Now and schedule updates

Basic scheduler

Cron expression

14

Days



Deployment settings

Deployment configuration

Choose from a list of default and custom deployment configurations. A deployment configuration is a set of rules that determines how fast an application is deployed and the success or failure conditions for a deployment.

CodeDeployDefault.AllAtOnce



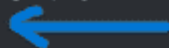
or

Create deployment configuration

Load balancer

Select a load balancer to manage incoming traffic during the deployment process. The load balancer blocks traffic from each instance while it's being deployed to and allows traffic to it again after the deployment succeeds.

☐ Enable load balancing



► Advanced - optional

Cancel

Create deployment group

- Finally Come on Add Deploy Stage and select that created Application name and Deployment group

Add deploy stage Info

Deploy - *optional*

Deploy provider

Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

AWS CodeDeploy

Region

US East (N. Virginia)

Application name

Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

ec2App

Deployment group

Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

AppDepGrp

Cancel

Previous

Skip deploy stage

Next

Step 17: Review And Create

- As a final step review and create it. By creating this we have successfully created a CI/CD pipeline in AWS.

aws

Services

Search

[Alt+S]

N. Virginia

Pratyush Sahoo

EC2

Developer Tools

CodePipeline

▶ Source • CodeCommit

▶ Artifacts • CodeArtifact

▶ Build • CodeBuild

▶ Deploy • CodeDeploy

▼ Pipeline • CodePipeline

Getting started

Pipelines

Pipeline

History

Settings

▶ Settings

Go to resource

Feedback

Source

Succeeded

Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415

Source

GitHub (Version 2)

Succeeded

- 22 minutes ago

928e84c6

928e84c6 Source: next commit

Disable transition

Build

Succeeded

Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415

Build

AWS CodeBuild

Succeeded

- 20 minutes ago

Details

View logs

928e84c6 Source: next commit

Disable transition

Deploy

Succeeded

Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415

Deploy

AWS CodeDeploy

✓

✓

✓